SEEDLINE · TRACEABILITY MATRIX

SeedLine traceability matrix

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/seedline		
Source	docs/03b-traceability.md @ 0a57933		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	Generated by .kit/trace.py

SeedLine traceability matrix

Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 17 user requirements, 18 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-001	PR-001	Single-seed placement	SDL-CAL-001, SDL-BLD-001, SDL-OPT-001	Analysis, cell speed against the fill limit (CAL section 3), TRL 3; sticky-belt bench test, TRL 4	At risk: cell speed 0.299 m/s at the 2.9 km/h rule against an assumed 0.30 m/s limit, no margin; misses and multiples cannot be calculated
UR-002	PR-002	Spacing uniformity	SDL-CAL-001, SDL-BLD-001	Analysis, Monte Carlo (CAL section 10), TRL 3; field count, TRL 5	At risk: 31 % of all spacings with the assumed miss and double rates; 9 % for single spacings
UR-003	PR-003	Spacing range	SDL-CAL-001, SDL-BLD-001	Analysis, spacing formula (CAL section 2), TRL 3	Met on paper with the ratio kit: 22 to 589 mm nominal, every target within 11.1 %; the base seeder (15 T only) covers 26 to 471 mm, within 17.3 %
UR-004	PR-004	Spacing follows travel	SDL-CAL-001, SDL-BLD-001	Analysis, traction estimate (CAL section 6a), TRL 3; field test, TRL 5	Met on paper: skid 1.2 to 1.7 %; rolling radius uncertain by up to 6.7 %, so calibrate per seedbed
UR-003, UR-005	PR-005	Seed size range	SDL-CAL-001	Analysis, cell tolerance rule (CAL section 4), TRL 3; bench test, TRL 4	Met on paper: 3.5 to 15 mm raw seed; pelleted carrot fits; raw 1 to 2 mm seed does not
UR-005	PR-006	Printable plates	SDL-CAL-001	Analysis, print estimate (CAL section 4), TRL 3; measurement, TRL 4	Met on paper: 1.39 h (maize) to 1.87 h (groundnut) at 20 % infill (SDL-DEC-001 F8), small margin for thick plates; accuracy to verify at TRL 4
UR-005	PR-007	Crop change	SDL-CAL-001, SDL-BLD-001	Inspection of the model, TRL 3; timed trial, TRL 4	To verify at TRL 4: side door and hand knob in the model; one 13 mm slot takes the 6, 8 and 10 mm plates with a printed liner pair
UR-006	PR-008	Sowing depth	SDL-CAL-001, SDL-BLD-001	Analysis, depth geometry (CAL section 9), TRL 3; field check, TRL 5	Met on paper: +/-10 mm holds for drive-wheel bumps within about +/-15 mm, taken as the ploughed-seedbed condition
UR-007	PR-009	Work rate	SDL-CAL-001	Analysis, field capacity (CAL section 8), TRL 3	Met on paper: 0.142 ha/h (7.1 h/ha) at 2.9 km/h
UR-008	PR-010	Push effort	SDL-CAL-001, SDL-BLD-001, SDL-STM-001	Analysis, planar statics with a soil model (CAL section 6), TRL 3; spring-scale test, TRL 4	At risk: 132 N horizontal pushing along the handle (206 N along it) in the design case; 179 N at best on a heavy, loose seedbed
UR-009	PR-011	Handling			Met on paper: 14.2 kg base, 14.8 kg with the marker kit (0.18 kg margin), with every fastener

User requirement	Product requirement	Title	Evidence	Verification	Status
			SDL-CAL-001, SDL-BLD-001	Analysis, mass build-up (CAL section 5), TRL 3; weighing, TRL 4	weighed from the detailed model and the mass fix of SDL-DEC-001 F6
UR-010, UR-017	PR-012	Hopper	SDL-CAL-001, SDL-BLD-001	Inspection, model volume (CAL section 8), TRL 3; timed trial, TRL 4	Met on paper: 2.42 L
UR-011	PR-013	Row-spacing kit	SDL-CAL-001, SDL-BLD-001	Inspection of the model, TRL 3	Met by design
UR-012	PR-014	Guarding	SDL-CAL-001, SDL-BLD-001	Inspection of the model, TRL 3	Met by design: printed shroud with an outboard face plate over both sprockets, the chain and the tensioner's full travel
UR-013	PR-015	Durability	SDL-CAL-001	Analysis, stress checks (CAL section 7), TRL 3; wear test, TRL 4	To verify at TRL 4: stresses are low, wear life unknown
UR-014	PR-016	Prototype cost	SDL-CAL-001	Analysis, priced BOM (`bom/bom.csv`), TRL 3	Met on paper: USD 266.10 estimated, USD 133.90 under the target
UR-015	PR-017	Replication cost	SDL-CAL-001	Analysis, priced BOM, TRL 3	Over target: USD 224.10 estimated, USD 24.10 over; USD 219.50 accepted for the prototype on 2026-10-02 (SDL-DDR-003 A3), and the plate liners and UV-stabilized door added USD 4.60 after it
UR-016	PR-018	Local build	SDL-CAL-001	Inspection of the model and build plan, TRL 3	Met by design

Gaps

None.